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Subject Activity

Semester/ Branch: - IV Sem/ CSE **Subject Code: - CS4T002**
Subject Name: - Design & Analysis of Algorithms

Title: Job Sequencing with Deadline – Algorithm Visualization

Objective:

The objective of this activity is to understand the **Job Sequencing with Deadline** algorithm by visualizing its execution.

- The algorithm aims to **maximize profit** by scheduling jobs within their respective deadlines.
- This problem is commonly solved using a **greedy approach**, ensuring jobs are scheduled in a way that yields the **highest possible earnings**.
- Through visualization, we will analyze:
  - How jobs are sorted based on profit.
  - How deadlines affect job selection.
  - The step-by-step execution of the scheduling process.
  - The impact of the algorithm's efficiency in terms of **time and space complexity**.

Activity Details

To practically understand the working of the Job Sequencing with Deadline algorithm, the following steps should be followed:

1. **Use Visualization Tools:**
   - Utilize platforms like **VisuAlgo**, **Algorithm Visualization**, or **custom Python/Java implementations with graphical output**.
   - Generate an **animated representation** of the algorithm's execution.
2. **Problem Statement:**
   - Given n jobs, each with a **profit** and a **deadline**, schedule jobs such that **maximum profit** is obtained without missing deadlines.

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- Each job takes **exactly one unit of time** to execute.

3. **Algorithm Execution Steps:**
   - **Step 1:** Sort all jobs **in descending order of profit**.
   - **Step 2:** Initialize an array (or timeline) to keep track of **available time slots**.
   - **Step 3:** Assign each job to the **latest available slot before its deadline**.
   - **Step 4:** Repeat until all jobs are scheduled or there are no free slots left.
   - **Step 5:** Compute and display the **total profit** achieved.
4. **Capture Screenshots:**
   - Take step-by-step snapshots showing:
     - Initial job list before sorting.
     - Sorted jobs based on profit.
     - Jobs being placed in available slots.
     - Final scheduled job sequence.
5. **Analyse Complexity:**
   - Determine **time complexity** based on sorting and scheduling steps.
   - Calculate **space complexity** based on additional storage used.

Introduction:

Problem Statement:

Given **N** jobs, where each job has:

- A **deadline** (before which it must be completed).
- A **profit** (earned if the job is completed within its deadline).

The objective is to schedule jobs in such a way that **total profit is maximized** while ensuring that **each job is scheduled within its deadline** and **only one job is executed at a time**.

Example with 7 Jobs

Given Data:

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Job	Deadline	Profit
Job1	4	20
Job2	1	10
Job3	1	40
Job4	1	30
Job5	3	50
Job6	2	60
Job7	2	70

Algorithm (Greedy Approach):

1. **Sort all jobs** in decreasing order of profit.
2. **Find the maximum deadline** among the given jobs.
3. **Initialize an array for scheduling jobs** (size = max deadline).
4. **Iterate through each job** in sorted order and assign it to the latest available slot before its deadline.
5. **If a slot is already occupied, move to an earlier slot**.
6. **If no slot is available, skip the job**.
7. **Return the scheduled jobs and total profit**.

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For this example, I'll first sort the jobs by profit in descending order, then schedule them using the **Job Sequencing with Deadline** algorithm. Finally, I'll generate visualizations to illustrate the process.

Step 1: Sort Jobs by Profit

Job	Deadline	Profit
Job7	3	70
Job6	2	60
Job5	3	50
Job3	1	40
Job4	1	30
Job1	4	20
Job2	1	10

Step 2: Job Scheduling Process

- The maximum deadline is **4**, so we have **4 available slots**.
- We iterate through the sorted job list and assign each job to the latest possible available slot.

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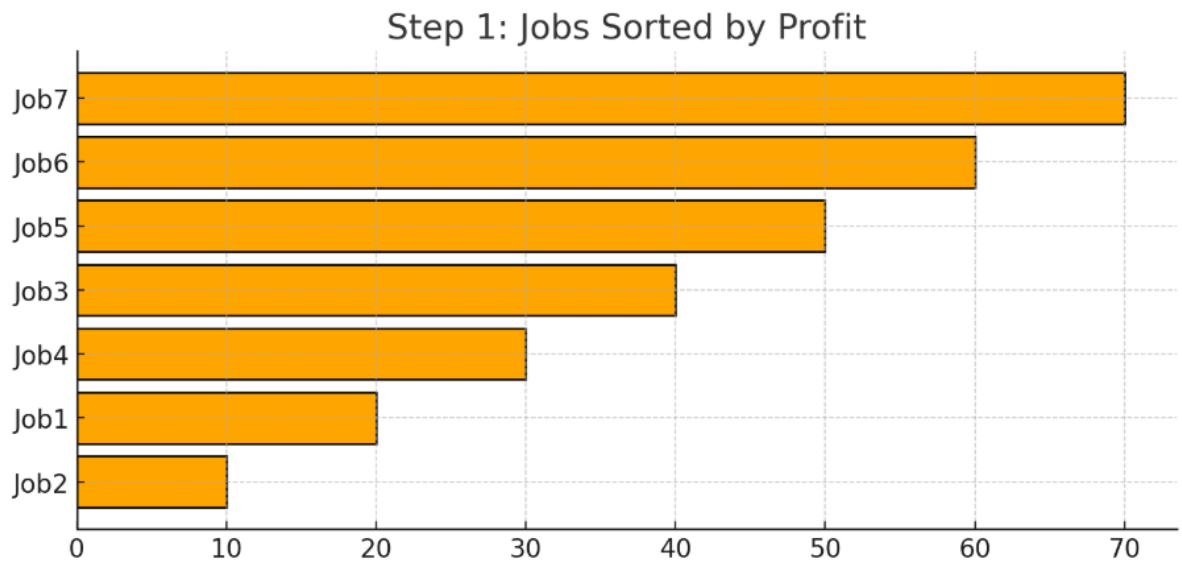
Time Slot	Assigned Jobs	Profit
Slot1	Job6	60
Slot2	Job7	70
Slot3	Job5	50
Slot4	Job1	20

Total Profit Earned:
70 + 60 + 50 + 20 = 200

Visual diagrams to represent these steps.



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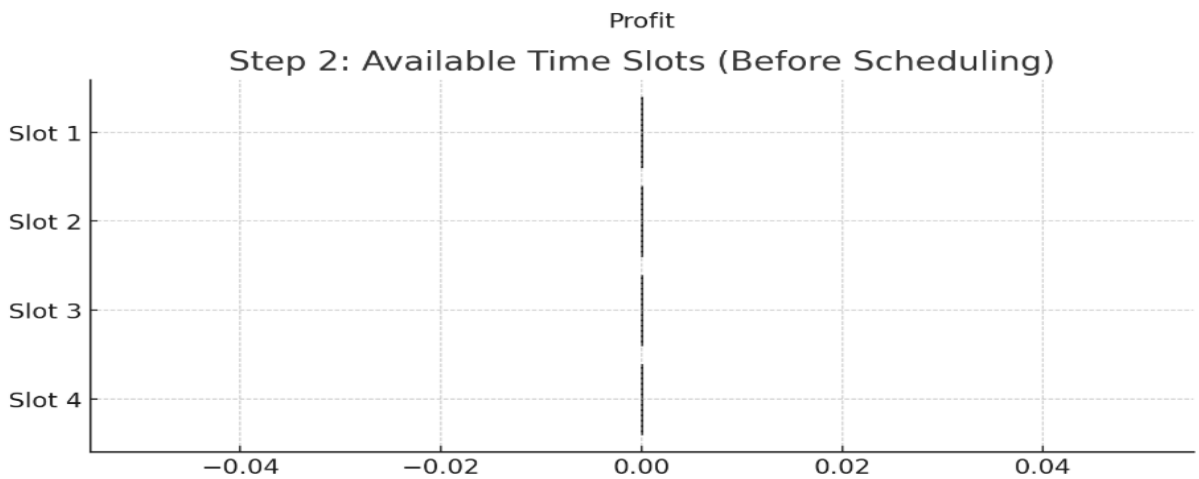

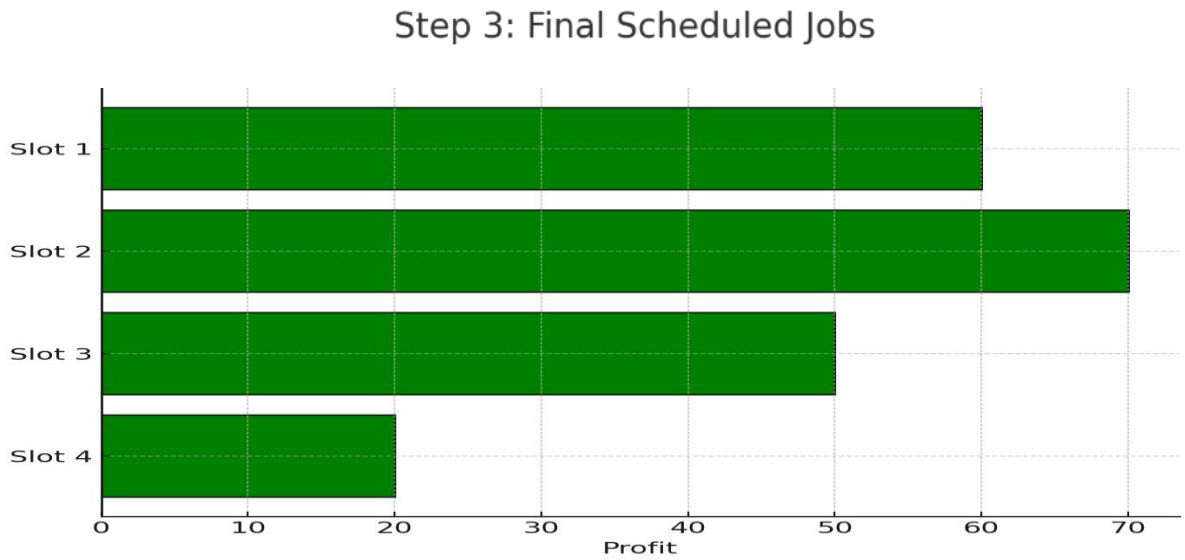

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Stepwise Procedure of Finding Time and Space Complexity:

Time Complexity Analysis:

1. **Sorting Jobs O (N log N))**
   - Jobs are sorted in descending order of profit using **Merge Sort or Quick Sort**, both of which have a time complexity of **O (N log N)**.
   - Sorting ensures that higher-profit jobs are scheduled first for maximum gain.
2. **Finding Available Slots & Scheduling Jobs O(N))**
   - Each job is placed in the **latest available slot** before its deadline.
   - Using an efficient **tracking method (e.g., Disjoint Set Union, DSU)**, slot assignment is done in **O(N)** instead of **(N^2)** (naïve approach).
3. **Overall Time Complexity: O (N log N)**
   - Sorting dominates the complexity, making the final time complexity **O (N log N)**.

Space Complexity Analysis:

1. **Storage for Jobs & Scheduling Array O(N)**
   - The algorithm stores job details **O(N)** and maintains a **schedule array** for tracking assigned slots **O(N)**.
2. **Total Space Complexity: O(N)**
   - Since we only use extra arrays proportional to the number of jobs, the final space complexity remains **O(N)**.

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Conclusion:

- The **optimal job sequence** for maximum profit is **Job6 → Job7 → Job5 → Job1**.
- **Total profit obtained: 200**.
- The **greedy algorithm ensures efficient job scheduling**, making it useful in **task scheduling, CPU job scheduling, and project management**.

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